

Clustering Algorithms – Quick Study Guide

K-Means

- Partition-based clustering algorithm.
- Goal: Minimize within-cluster sum of squared distances.
- Steps:
 1. Choose number of clusters K.
 2. Initialize K centroids randomly.
 3. Assign each data point to nearest centroid.
 4. Recompute centroids.
 5. Repeat until centroids stabilize.
- Objective Function: minimize $\sum ||x - \mu||^2$

Hierarchical Clustering

- Builds clusters in a tree structure (dendrogram).
- Two types: Agglomerative (bottom-up) and Divisive (top-down).
- Steps (Agglomerative):
 1. Start with each data point as its own cluster.
 2. Compute distance between clusters.
 3. Merge closest clusters.
 4. Repeat until one cluster remains.

DBSCAN

- Density-based clustering algorithm.
- Clusters formed based on dense regions.
- Key parameters: epsilon (ϵ) and minPts.
- Point types: Core point, Border point, Noise point.
- Advantages: Detects arbitrary shapes and handles noise.

Mean Shift

- Density-based algorithm using kernel density estimation.
- Steps:
 1. Place a window around each data point.
 2. Compute mean of points in window.
 3. Shift window to mean location.

- 4. Repeat until convergence.

Gaussian Mixture Model (GMM)

- Probabilistic clustering method.
- Assumes data generated from mixture of Gaussian distributions.
- Uses Expectation Maximization (EM).
- Steps:
 1. Initialize parameters.
 2. Expectation step: compute probability of cluster membership.
 3. Maximization step: update Gaussian parameters.

Spectral Clustering

- Graph-based clustering method.
- Uses eigenvectors of similarity matrix.
- Steps:
 1. Construct similarity matrix.
 2. Compute Laplacian matrix.
 3. Compute eigenvectors.
 4. Apply K-means on eigenvectors.

BIRCH

- Balanced Iterative Reducing and Clustering using Hierarchies.
- Efficient for large datasets.
- Builds clustering feature tree (CF Tree).

Affinity Propagation

- Message passing algorithm.
- Finds exemplars automatically.
- No need to predefine number of clusters.

OPTICS

- Ordering Points To Identify Clustering Structure.
- Extension of DBSCAN.
- Handles varying density clusters.